Interactive House Price Explanation Dashboard using LIME

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***Abstract*—Machine-learning models for house-price prediction often achieve strong predictive performance but remain difficult for end users to interpret due to their black-box nature. This lack of transparency can reduce trust and limit practical adoption in real-estate decision making. In this project, we present an interactive web-based dashboard that combines house-price prediction with explainable artificial intelligence techniques. A regression model is trained on a tabular housing dataset, and local explanations are generated using LIME to highlight feature-level contributions for individual predictions. The dashboard integrates data exploration, geospatial visualization, and a real-time "what-if" analysis that allows users to modify input features and immediately observe changes in predictions and explanations. Experimental results demonstrate that the system provides intuitive and informative explanations while preserving predictive functionality. By unifying prediction, explanation, and visualization in a single interactive interface, the proposed approach improves transparency and user understanding of house-price prediction models.**

I. Introduction

Predictive models for house prices are widely used in real-estate analytics, but they are often presented as opaque black-box systems. Users, such as home buyers or market analysts, frequently lack insight into *why* a model assigns a particular price. This undermines trust, interpretability, and the practical usability of machine-learning systems in decision making.

Our project addresses this problem by building an interactive dashboard that explains individual model predictions using LIME. The seminal work by Ribeiro et al. [1] introduced LIME as a local, model-agnostic explanation technique that approximates complex models with simple interpretable surrogates. LIME has been widely adopted in explainable AI, and many subsequent works build upon it, including extensions such as SP-LIME (for selecting representative global examples), Anchors (stability-focused rule explanations), and SHAP (a Shapley-value-based alternative). These follow-up works emphasize two core challenges of explainability: fidelity to the underlying model and stability of the explanation.

We focus on applying these ideas to the domain of house-price prediction. The dataset used in this project follows the preprocessing pipeline commonly demonstrated in introductory machine-learning tutorials, such as [2]. Traditional hedonic regression assumes linear relationships between features and price, but modern ML models can capture nonlinear interactions that are difficult for users to interpret without an explanation layer. By integrating LIME and interactive visualization, we extend the original LIME framework into a user-centered tool that supports real-time feature manipulation and intuitive interpretation of model behaviour.

II. Related Work

House price prediction has been a recurring topic in machine learning and real estate analytics. Many introductory and practical studies focus on tabular regression pipelines using conventional models such as linear regression, decision trees, or ensemble methods. A widely circulated example is the tutorial by GeeksforGeeks [2], which demonstrates a standard preprocessing workflow and simple regression modeling. While such approaches are accessible, they provide no explanation layer, making the resulting predictions difficult for end users to interpret.

The survey-style paper "House Price Prediction using Machine Learning in Python" (Kalra et al., 2021) summarizes basic predictive techniques for real-estate valuation and has been used as a reference point in several later works. Its citations include: (1) Srivastava et al. (2023), who revisit house-price prediction using updated machine-learning pipelines; (2) Rathod et al. (2023), who apply ML methods to accident detection but cite Kalra et al. as an example of structured tabular prediction methodology; (3) Titarmare et al. (2023), who use similar supervised-learning techniques in an interactive voice-assistant application; and (4) Dehankar et al. (2023), who employ tabular ML models for human-state detection. Although the cited works span unrelated domains, they reflect a shared reliance on simple, non-interpretable prediction models.

The references of Kalra et al. also include several classical contributions. Shiller (2007) provides an economic analysis of house-price trends and market behaviour. Adair et al. (1996) and related hedonic-pricing literature emphasize interpretable linear relationships between housing attributes and market value, forming the conceptual foundation for traditional valuation models. Additional methodological references include early discussions of stochastic gradient descent for linear regression and broader data-mining overviews, both illustrating the shift from econometric modelling toward modern ML approaches.

However, while these works focus on prediction accuracy or economic interpretation, they do not provide user-centered explanations of model behaviour. This gap motivates the use of explainable artificial intelligence (XAI).

The seminal contribution in this area is LIME, introduced by Ribeiro et al. [1]. LIME approximates black-box predictions with sparse, locally faithful surrogate models, enabling users to inspect feature-level contributions for individual instances. The paper also proposes SP-LIME, a procedure for selecting representative examples to convey global model behaviour. LIME has inspired numerous follow-ups addressing issues such as explanation stability, global consistency, and user trust. Despite its wide adoption, LIME is typically used in a static, Python-notebook-style workflow rather than an interactive exploratory interface.

Our project extends this line of work by integrating LIME explanations into an interactive dashboard specifically designed for house-price prediction. In contrast to previous studies, which either emphasize predictive accuracy or static explanation plots, we focus on dynamic visual explanations, user manipulability (what-if feature adjustment), and improved interpretability for non-expert users.

III. Methods

Our system consists of four main components: data exploration, a regression model for predicting house prices, an explanation module based on LIME, and an interactive web interface for visualizing explanations and data exploration.

A. Data Exploration

Before model training, we perform an in-depth exploration of the dataset to understand distributions, relationships, and spatial patterns. This includes:

- **Price distribution:** Histograms to visualize how median house prices are distributed and identify common price ranges.
- **Price vs. Median Income:** Scatter plots showing the relationship between median income and house prices, with population density encoded in color.
- **Geospatial distribution:** Mapping house prices across California to detect regional patterns and clusters.
- **Feature correlations:** Heatmaps illustrating pairwise correlations between features to identify potential multicollinearity or influential predictors.

B. Data Preprocessing

The dataset is preprocessed using standard machine-learning procedures: missing-value handling, one-hot encoding of categorical attributes, and normalization where appropriate. We split the dataset into an 80/20 train–test partition.

C. Model Training and Explanation

We train a predictive model (e.g., linear regression, random forest, or gradient-boosted trees). The chosen model exposes a prediction function that LIME queries to construct local explanations. For each selected instance, LIME generates perturbed samples around the input point, queries the black-box predictor, and fits a sparse linear model that approximates the local decision surface. The resulting feature weights indicate how each attribute contributes to increasing or decreasing the predicted price.

D. Interactive Dashboard

Explanations and data exploration are rendered in a web dashboard, which includes:

- A local explanation view showing the top positive and negative feature influences.
- A global representative overview using example instances.
- A real-time "what-if" simulator allowing users to change feature values and observe prediction and explanation updates.
- Visualizations from the data exploration phase, enabling users to interactively explore distributions, relationships, correlations, and geospatial patterns.

IV. Results

The developed system successfully combines predictive modeling, explainable artificial intelligence, and interactive visualization into a single web-based dashboard. The regression model produces plausible house-price predictions across the full range of feature values present in the dataset. When users modify input features in the sidebar, the predicted price updates immediately, enabling real-time exploration of the model's behavior.

Local explanations generated by LIME provide clear insight into individual predictions. For each selected instance, the explanation highlights the most influential features contributing positively or negatively to the predicted house price. In most cases, median income and geographic location (latitude and longitude) emerge as the strongest contributors, which aligns with known characteristics of the California housing market. The explanations remain intuitive and interpretable, as they are presented as simple feature-weight contributions rather than abstract model internals.

The interactive "what-if" analysis further enhances interpretability. By varying a single feature while keeping others fixed, users can observe how changes affect the predicted price. This allows users to reason about marginal effects and nonlinear relationships that are not obvious from raw model output alone.

In addition to local explanations, the data exploration module provides valuable global context. Histograms reveal the skewed distribution of house prices, scatter plots illustrate the strong correlation between income and price, and the correlation heatmap exposes dependencies among features. The geospatial visualization highlights regional clustering of house prices, with higher values concentrated along coastal areas, which is consistent with established economic observations.

Overall, the results demonstrate that integrating LIME explanations with interactive visualization significantly improves transparency and user understanding of house-price prediction models, without sacrificing predictive functionality.

V. Conclusion

In this project, we presented an interactive dashboard for house-price prediction that integrates explainable artificial intelligence techniques with intuitive visual analytics. By combining a machine-learning regression model, LIME-based local explanations, and interactive data exploration, the system addresses the common lack of transparency associated with black-box predictive models.

A. Summary of Contributions

Unlike traditional static prediction pipelines, the proposed dashboard enables users to actively explore model behavior. Local explanations clarify why a specific prediction was made, while the "what-if" analysis allows users to test hypothetical scenarios and observe how individual features influence the outcome. The inclusion of data exploration and geospatial visualization further grounds model predictions in the underlying dataset, helping users develop trust and contextual understanding.

B. Limitations

One limitation of the current system is that the predictive model is trained only on the feature ranges present in the dataset. When users explore values outside this range using the what-if simulator, predictions may become unreliable, as the model has not been exposed to such examples during training. This limitation is especially relevant for tree-based models, which generally do not extrapolate well beyond the observed data.

C. Future Work

As future work, the project could be extended by incorporating additional datasets, particularly housing data from Slovenia. This would enable region-specific analysis and improve the practical relevance of the system in a local context. However, such an extension would require training the predictive model on a sufficiently large and representative Slovenian housing dataset. A small subset of local data could not be directly integrated into the current model without retraining, as the underlying feature distributions and market dynamics differ significantly across regions.

In practice, acquiring large-scale Slovenian housing data presents nontrivial challenges. Although platforms such as *Nepremicnine.net* contain extensive listings, access to such data is limited and large-scale collection is not straightforward. A robust solution would require the development of an automated data ingestion pipeline capable of extracting and updating listings at scale, while respecting website access constraints and usage policies. This could involve the careful design of a crawling or data-collection mechanism with appropriate rate limiting and compliance considerations.

Furthermore, the visualization component could be enhanced with alternative color palettes designed for colorblind users, improving accessibility and inclusiveness. Additional improvements could include uncertainty-aware predictions and explicit handling of out-of-range inputs. Together, these extensions would further strengthen the usability, robustness, and real-world applicability of the dashboard in decision-making scenarios.

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